L2 : MORE ABOUT FACTORIZATION (B)

Checkpoint 1

$a^2 + 2ab + b^2 = (a+b)^2$

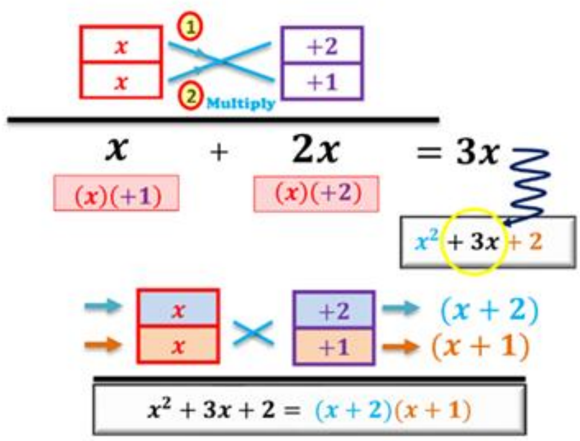
Checkpoint 2

$a^2 - b^2 = (a+b)(a-b)$

Checkpoint 3

Factorize $x^2 + 3x + 2$

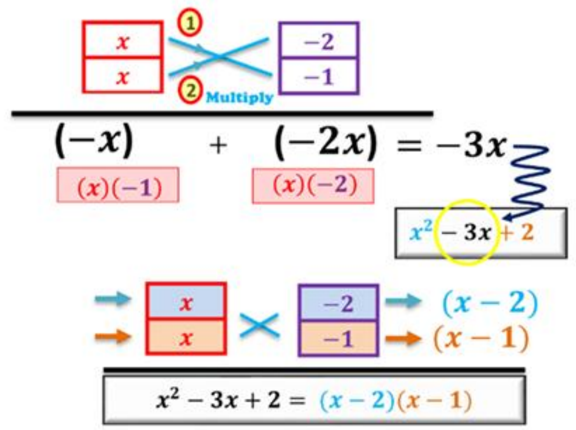
x^2	2
x	$+2$
x	$+1$



Checkpoint 4

Factorize $x^2 - 3x + 2$

x^2	2	
x	$+2$	-2
x	$+1$	-1



Checkpoint 5

Factorize $x^2 + 2x - 3$

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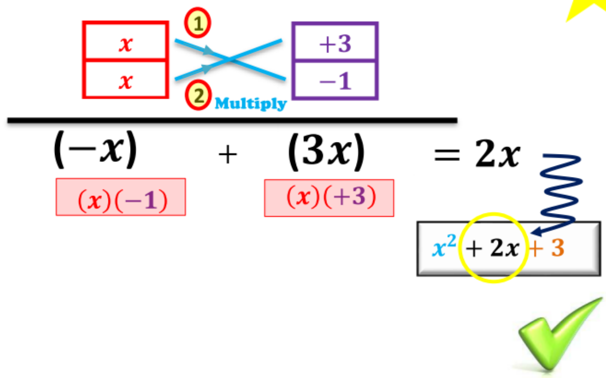
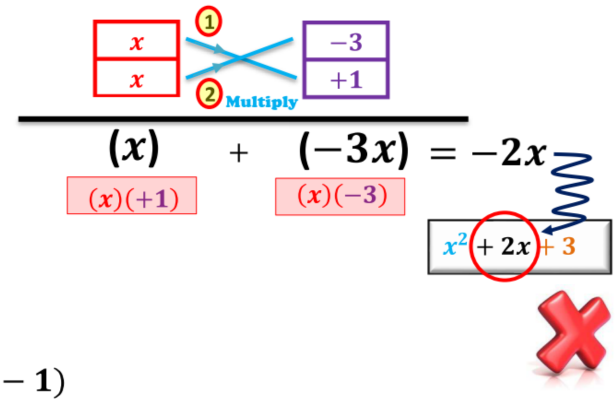
Trial 1

x^2	-3	
x	$+3$	-3
x	-1	$+1$



Trial 2

x^2	-3	
x	$+3$	-3
x	-1	$+1$



Final Ans:

$x^2 + 2x - 3 = (x+3)(x-1)$

Let's get started!

Factorize the following expressions.

1. $a^2 + 24a + 95$

2. $b^2 - 48b + 176$

3. $c^2 - 13c - 90$

4. $d^2 + 17d - 200$

5. $66 + w^2 + 17w$

6. $-6s + s^2 - 91$

7. $-32 - 4h + h^2$

8. $-42 + 11r + r^2$

9. $15p - p^2 - 44$

10. $-17n - n^2 - 72$

11. $2s^2 - 26s + 72$

12. $3t^2 - 12t - 63$

PRE S3 MATHS L2 –
MORE ABOUT FACTORIZATION (B)

Factorize the following expressions.

13. $7x^2 - 10x + 3$

14. $5y^2 - 11y + 2$

15. $4c^2 + 9c + 2$

16. $4s^2 - 11s + 6$

17. $2a^2 - 5ab + 2b^2$

18. $3p^2 - 8pq + 5q^2$

19. $5f^2 + 24fg + 16g^2$

20. $6s^2 + 17st + 12t^2$.

21. (a) $2s^2 - 11s + 14$
    (b) $(s-2)^2 + 2s^2 - 11s + 14$

22. (a) $3a^2 - 7ac + 2c^2$
    (b) $3a^2 - 7ac + 2c^2 - ab + 2bc$

23. (a) $7m^2 + 16mn + 4n^2$
    (b) $7m^2 + 16mn + 4n^2 + 5m + 10n$

24. (a) $x^2 - 8x + 12$
    (b) $xy - x^2 - 6y + 8x - 12$

Factorize the following expressions.

25. $4a^2 + 16a - 9$

26. $24b^2 - b - 10$

27. $22c^2 - 67c + 30$

28. $9d^2 - 4d - 5$

29. $21a^2 - 8ab - 4b^2$

30. $4m^2 + 4mn - 15n^2$

31. $10p^2 + 9pq - 7q^2$

32. $12u^2 - 7uv - 12v^2$

33. $7m^3n - 41m^2n^2 - 6mn^3$

34. $60b^2 - 24b^4 + 4b^3$

35. (a) $18x^2 - 3xy - 10y^2$
(b) $24x - 20y - 18x^2 + 3xy + 10y^2$

36. (a) $x^2y - 4xy^2 - 32y^3$
(b) $x^2y + 2x^2 + 16xy - 4xy^2 + 32y^2 - 32y^3$

PRE S3 MATHS L2 –
MORE ABOUT FACTORIZATION (B)

Factorize the following expressions.

37. $30m^2-6(12m+9)$

38. $5n(2n-7)-20$.

39. $(9p-12)(6p+6)-15p$

40. $9c-6(c+1)(5c-2)$

41. $(5p+8r)(3p-10r)-3p(p-8r)$

42. $(x+3)(2x-1)+3(x-9)(x+1)$

43. $(y+5)^2+3(y+4)-1$

44. $(z-6)^2+2(z-3)-6$

45. $(x+3)^3-25x-75$

46. $36y-144-(y-4)^3$

47. Factorize
   (a) $3x-9y$,
   (b) $x^2+2xy-15y^2$,
   (c) $x^2+2xy-15y^2-3x+9y$

48. Factorize
   (a) $13s+52t$,
   (b) $s^2+3st-4t^2$,
   (c) $s^2+3st-4t^2+13s+52t$.

49. Factorize
   (a) uv^2-v^2,
   (b) u^2-6u+5,
   (c) $uv^2-v^2+u^2-6u+5$.

50. Factorize
   (a) $-3ab^2-3b^2$,
   (b) a^2+5a+4,
   (c) $a^2+5a+4-3ab^2-3b^2$.

51. Factorize
   (a) $25p^2-16$,
   (b) $5p^2q-11pq-12q$,
   (c) $25p^2-16-5p^2q+11pq+12q$.

52. Factorize
   (a) $14x^2-25x+6$,
   (b) $49x^2-4$,
   (c) $(42x^3-75x^2+18x)-(98x^2-8)$.

Unit Education

Final Boss!

53. (a) Factorize $2x^2 - x - 10$.
    (b) Hence, factorize
        (i) $2(y+2)^2 - (y+2) - 10$,
        (ii) $2(2z-5)^2 - 2z - 5$.

54. (a) Factorize $a^2 - 7a - 30$.
    (b) Hence, factorize
        (i) $(b-5)^2 - 7(b-5) - 30$,
        (ii) $(3c-2)^2 - 7(3c-2) - 30$.

55. (a) Factorize $12m^2 - 35m - 3$.
    (b) Hence, factorize
        $24s(4t-5)^2 - 70s(4t-5) - 6s$.

56. (a) Factorize $5x^2 - 17xy + 6y^2$.
    (b) Hence, factorize
        $45(p-q)^2 - 51(p^2 - q^2) + 6(p+q)^2$

57. (a) Factorize $p^2 - 13p + 36$.
    (b) Hence, factorize $q^4 - 13q^2 + 36$.

58. (a) Factorize $b^2 - 14b + 40$.
    (b) Hence, factorize $(a^2 - 3a)^2 - 14(a^2 - 3a) + 40$.

59. (a) Factorize $5a^2 - 31a + 6$.
    (b) Hence, factorize $5b^3 - 31b^2 + 51b - 9$.

60. (a) Factorize $20y^2 - 49y + 30$.
    (b) Hence, factorize $4y^3 + 15y^2 - 49y + 30$.

61. (a) Factorize $6x^2 - 7x - 10$.
    (b) Hence, simplify $\dfrac{6x^2 - 7x - 10}{4x - 8} \times \dfrac{2}{6x + 5}$.

62. (a) Factorize
    (i) $9x^2 - 16y^2$,
    (ii) $15x^2 - xy - 28y^2$.
    (b) Hence, simplify $\dfrac{15x^2 - xy - 28y^2}{45x - 63y} \div \dfrac{9x^2 - 16y^2}{15x - 6y}$.

63. (a) Factorize $x^2 + 5x - 24$.
    (b) Hence, simplify $\dfrac{x}{x^2 + 5x - 24} - \dfrac{1}{x + 8}$.

64. (a) Factorize $3x^2 - 13x - 10$.
    (b) Hence, simplify $\dfrac{1}{x - 5} - \dfrac{2x + 7}{3x^2 - 13x - 10}$.